

SAMADHAN DAYANAND MANE

AI Engineer | Conversational AI & LLM Integration | Python Developer

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PROFESSIONAL SUMMARY

AI/ML Engineer and final year Computer Engineering student specializing in production grade conversational AI and LLM powered systems. Built and deployed 3 end to end AI applications including a multi tenant due diligence platform with agentic RAG that secured 1st place at Neobim Hackathon (₹1.8L prize). Expertise in LLM orchestration (LangChain, LangGraph, OpenAI, Gemini, Groq, Mistral), vector retrieval (Pinecone, ChromaDB), Fine Tuning and containerized deployment (Docker, AWS).

TECHNICAL SKILLS

AI & ML: Machine Learning, Deep Learning, NLP, Transformers, Neural Networks, Hugging Face, RAG, Chatbot, Agentic AI, LLM Integration, Prompt Engineering, Fine Tuning

Programming Languages: Python, Java, SQL, JavaScript, C++

Frameworks and Libraries: TensorFlow, Keras, NumPy, Pandas, Scikit learn, LangChain, LangGraph

Data & Vector Databases: Pinecone, ChromaDB, MongoDB Atlas

Cloud & Tools: AWS, Docker, Git, GitHub

PROJECTS

Abyss-ai-dd: Multi-Tenant AI Due Diligence Platform

Mar–Apr 2026

Team of 4 | FastAPI, React, TypeScript, Pinecone, Groq, Docker / github.com/samadhanmane/Abyss-ai | Live: abyss-ai-gray.vercel.app

- Architected an **8-step agentic RAG state machine** (Rewrite → Retrieve → Completeness → Think → Audit → Structure) that enforces source-grounded, citation-backed due diligence analysis, surpassing simple retrieval-augmented generation.
- Built a **self-improving feedback loop** that ingests human reviewer corrections directly into the system prompt, improving answer relevance by 23% without model retraining.
- Engineered a **fault tolerant ingestion pipeline** for 15+ file types using PyMuPDF/Tesseract OCR, achieving 94% retrieval recall through semantic chunking and hybrid embeddings indexed in Pinecone.
- Orchestrated a **full stack, multi tenant platform** with 11 core modules (Auth, DMS, Finance, Audit), deployed on AWS, with route based code splitting reducing frontend bundle size by 63%.

MOM-ai: AI Meeting Research Assistant

Python, LangChain, Mistral AI, ChromaDB, Whisper, Streamlit / github.com/samadhanmane/MOM-ai | Live: [MOM — AI Video Assistant · Streamlit](https://mom-ai-video-assistant.streamlit.app)

- Built an end to end AI pipeline that converts audio, and YouTube recordings across 7 formats into structured executive summaries, action items, key decisions, and open questions with 92% accuracy on real meeting transcripts.
- Integrated dual speech to text engines (OpenAI Whisper for English, Sarvam AI for Hinglish) enabling robust multilingual transcription that handles both global and India-specific language patterns effectively.
- Built a conversational RAG layer with LangChain over a ChromaDB vector store and Mistral embeddings, letting users ask natural language questions directly against their own meeting transcripts conversationally and receive contextual, citation backed responses.

Multi-Tenant College Management Platform: AI Booking Chatbot

React, Node.js/Express.js, MongoDB Atlas, Gemini API / github.com/samadhanmane/BookMyHall | Live: book-my-hall.vercel.app

- Engineered an **enterprise multi-tenant facility booking system** with a **transactional Gemini AI agent** that executes natural-language-driven slot bookings, including temporary concurrency locking to prevent double-booking
- Designed a **role based access control (RBAC) system** with 10 permission tiers (Super Admin to Technician) and **isolated data per organization**, enabling secure deployment for multiple colleges on a single platform.
- Architected **multi-stage approval workflows** for canteen requisitions (Faculty → HOD → Registrar → Director) and maintenance ticketing (HOD → Workshop HOD → Technician), with real-time status tracking and email notifications.
- Built a **monorepo full stack application** with RESTAPI backend, MongoDB, and React, implementing **5-minute slot locks, race condition prevention, and asynchronous email queues** for production readiness.

EDUCATION

MIT Academy of Engineering, Pune: B.Tech, Computer Engineering

2023–2027 | CGPA: 7.49/10

ACHIEVEMENTS

- Winner, Neobim Hackathon (2026): 1st place (₹1.8L prize pool) for ABYSS, an AI powered due diligence platform recognized for its novel agentic architecture and production readiness.